

NAME

observatory -- Examine and set observatory parameters

USAGE

observatory command obsid [images]

PARAMETERS**command = "list" (set|list|images)**

Command option which is one of "set", "list", or "images". The set command sets the default observatory task parameters for the specified observatory. The list command lists the observatory parameters for the specified observatory but does not modify the task parameters. The images command lists the observatory parameters for a list of images. The list and images commands examine and verify the observatory parameters applied by other tasks using the observatory database facility.

obsid = "?"

Observatory identification to be set, listed, or used as the default for images without the OBSERVAT keyword. The observatory ID is one of those in the database (case ignored), the special string "observatory" to default to the environment variable "observatory" or the *observatory.observatory* parameter, "obspars" to select the parameters in the **observatory** task, or "?" to list the observatories defined in the database.

images

List of images to be examined with the "images" command. The images are checked for the OBSERVAT keyword to determine the observatory parameters to be listed, otherwise the observatory given by *obsid* is used.

verbose = no

Verbose output? Because there are a number of different ways in which observatory information is determined this option prints detailed information on how the observatory database and parameters are ultimately selected.

observatory

The default observatory used by tasks which use the special observatory identification "observatory". The value is one of the observatory names in the observatory database (case ignored) or the special value "obspars" to select the parameters defined in this task. There is no default to force users to set it at least once.

name Observatory name.

longitude

Observatory longitude given in degrees west.

latitude

Observatory latitude in degrees. Positive latitudes are north and negative latitudes are south.

altitude

Observatory altitude in meters above sea level.

timezone

Observatory time zone. The time zone is the number of hours west of Greenwich or the number of hours to be added to local time to obtain Greenwich time.

ENVIRONMENT VARIABLES**obsdb**

This variable selects the observatory database. If not defined it to noao\$lib/obsdb.dat.

observatory

This variable selects the observatory entry whenever a observatory name "observatory". If not defined the value of the task parameter *observatory.observatory* is used.

IMAGE HEADER KEYWORDS

The observatory identification for images is first sought under the image header keyword OBSERVAT. This always takes precedence over any other means of defining the observatory.

DESCRIPTION**OBSERVATORY PARAMETERS IN THE NOAO PACKAGE**

Some astronomical data reduction and analysis tasks perform computations requiring information about where the data was observed. For example a number of **noao** tasks make corrections for the airmass. Generally they look for an airmass in the image header and if it is not present they attempt to compute it from other image header parameters. The information about time and telescope coordinates of the observation are often in the image header but the observatory latitude is not. The task must get this information somehow.

Prior to IRAF V2.10 tasks generally had explicit parameters, such as latitude, with default values pointing (using parameter redirection) to the parameter of the same name in the **observatory** task. The user was required to know the values of the observatory parameters and manually change them for data from different observatories. In V2.10 an observatory database has been implemented. Observatory parameters are stored in a simple text file and tasks obtain observatory related parameters by specifying an observatory identification.

In general the information about the observatory should be directly associated with the image data. Unless stated otherwise in the description of a task, tasks which require observatory information will first look for the image header keyword OBSERVAT. The value of this keyword is the observatory identification used to index the observatory database. The task will then look up any observatory parameters it needs in the observatory database. Data from observatories that support this keyword will, therefore, always use the correct observatory parameters without user intervention. All observatories which export FITS image data are urged to adopt the OBSERVAT keyword (a keyword recommended by the FITS standard).

For image data which do not identify the observatory in this way and in tasks which do not operate on images (such as astronomical calculator tools), the observatory must be specified by the user. Most tasks provide an "observatory" parameter which either directly selects the observatory or use special values for defining the observatory with an environment variable or the parameters from the **observatory** task.

An observatory is specified by the identification name used in the observatory database. The names in the database may be listed using the **observatory** task as described below. If the desired observatory is not in the database a user may copy/create their own database and select it with the environment variable "obsdb", modify the standard database if allowed (any changes to the distributed version should be forwarded to iraf\$noao.edu), or use the special observatory name "obspar.s". The last option directly uses

the parameters in the **observatory** task which can be set to any values using the normal parameter editing mechanism.

The default value for the observatory parameter in a "observatory". This special name directs the task to look first for the environment variable of the same name and then at the *observatory* parameter of the **observatory** task. The environment variable allows users or sites to set the default observatory in their login files and site defaults. Also it is simple to change the default observatory either with a **reset** command or the **observatory** command.

The observatory database is selected by the environment variable "obsdb". The default when the variable is not defined is the **noao** package library database file "noao\$lib/obsdb.dat". The use of an environment variable allows users to permanently change the default database in the OS environment (when IRAF has access to it such as in UNIX systems) or in the startup IRAF environment as set in the "login.cl" or "loginuser.cl" files. One can, of course, change it during a session with the set or reset commands. For sites which want to customize the observatory mechanism the environment variables can also be set and changed in the files "hlib\$zzsetenv.def", "noao\$lib/zzsetenv.def", and the template login file "hlib\$login.cl".

An observatory database file consist of a simple list of keyword=value pairs with arbitrary whitespace allowed. An observatory entry begins with the observatory keyword and extends to the next observatory keyword or the end of the file. The observatory identification should be the same as the string used in the OBSERVAT image header parameter for data from that observatory. The default file noao\$lib/obsdb.dat begins as follows:

```
# Observatory Parameters. Taken from the Almanac.
#
# Observatories wishing to be added or make changes in the default
# distributed database should send information to iraf@noao.edu.

observatory = "kpno"
    name = "Kitt Peak National Observatory"
    longitude = 111:36.0
    latitude = 31:58.8
    altitude = 2120.
    timezone = 7

observatory = "ctio"
    <etc>
```

In summary, access to observatory parameters is now done by referencing the image header keyword OBSERVAT and, if not defined, determine the observatory name from a task parameter. The environment variables "observatory" and "obsdb" can be set by the user to select alternate observatories and observatory database files. For data without an observatory entry the observatory can be set to "obspars" or the user may make their own observatory database.

THE OBSERVATORY TASK

The **observatory** task serves a number of functions. It examine the observatory database, verify the observatory parameters which will be used by other tasks, particularly those operating on images, set the default observatory if not defined by other means, set observatory parameters explicitly, especially when there is no observatory database entry, and as a parameter set for tasks which explicitly reference observatory parameters. The **verbose** parameter also provides a detailed check of the steps used to determine the observatory database, observatory identification, and observatory parameters.

The *command* parameter takes the values "set", "list", or "images". The *obsid* parameter supplies the observatory identification and the *images* parameter is used to specify a command. The parameters are query parameters and so may be either queried or simply typed on the command line.

The "set" command prints the observatory parameters for the specified observatory and sets many of these in the **observatory** task parameters. This command is used to set the default observatory parameters for tasks where images are not used, the images do not contain the observatory id, or direct references to specific parameters with parameter redirection (for example ")observatory.latitude") are used.

The "list" command is similar to the "set" command except the task parameters are not modified. It is used to list observatory parameters. It is also use with the special observatory identifications to list the entries in an observatory database and verify the observatory to be used by tasks which do not operate on images. The special value "?" lists the entries in the database. The special value "observatory" lists the observatory defined by the "observatory" environment variable or that given by the *observatory.observatory* parameter. The special value "obsvars" simply lists the observatory task parameters.

The "images" command lists the observatory information applicable to one or more images. In particular, the observatory identification is first sought in OBSERVAT image header keyword and, if not found, the *obsid* parameter is used. Often the default observatory is "observatory" to follow the same search path used by other tasks.

The *verbose* parameter prints additional detailed information. It prints the database used and whether it is selected by default (noao\$lib/obsdb.dat) or by the "obsdb" environment variable. When the observatory is defined as "observatory" it indicates whether the observatory is defined by the environment variable "observatory" or by the observatory task. When listing images it prints the OBSERVAT keyword or the default observatory assigned.

For observatories not in a database the name, latitude, longitude, altitude, and time zone parameters may be set using **eparam**. The observatory id must be set to "obsvars" in this case. These parameters will then be referenced by other tasks in which the observatory is specified as "obsvars". This allows arbitrary observatory parameters to be set without creating or modifying an observatory database. However, it is advisable to create a local database and also send the observatory information to the IRAF group at NOAO for inclusion in the default database.

EXAMPLES

1. List the observatory entries in the database:

```
cl> observatory list ? v+
Using default observatory database: noao$lib/obsdb.dat
```

```
default: Kitt Peak National Observatory
kpno: Kitt Peak National Observatory
ctio: Cerro Tololo Interamerican Observatory
eso: European Southern Observatory
lick: Lick Observatory
mmt: Whipple Observatory
cfht: Canada-France-Hawaii Telescope
lapalma: Roque de los Mucachos, La Palma
```

2. Set the observatory parameters for Cerro Tololo:

```
cl> observatory set ctio
Observatory parameters for Cerro Tololo...
  observatory = ctio
  timezone = 5
  altitude = 2215.
  latitude = -30:09.9
  longitude = 70:48.9
  name = 'Cerro Tololo Interamerican Observatory'
cl> lpar observatory
```

command = "set"	Command (set list images)
argument = ctio	Observatory or images
(observatory = "ctio")	Observatory identification
(name = "Cerro Tololo...")	Observatory name
(longitude = 70.815)	Observatory longitude (degrees)
(latitude = -30.165)	Observatory latitude (degrees)
(altitude = 2215.)	Observatory altitude (meters)
(timezone = 4)	Observatory time zone
(verbose = no)	Verbose output?
(mode = "q")	

3. Set the observatory parameters to use the environment variable "observatory" and verify it.

```
cl> set observatory=cfht
cl> observatory list observatory
Observatory parameters for Canada-France-Hawaii Telescope
  observatory = cfht
  timezone = 10
  altitude = 4215
  latitude = 19:49.6
  longitude = 155:28.3
  name = 'Canada-France-Hawaii Telescope'
```

4. Change the default observatory database and verify verbosely:

```
cl> set observatory="sco"
cl> set obsdb="/local/iraf/obsdb.dat"
cl> type obsdb$
# Local Observatory Parameters.

observatory = "sco"
  name = "Small College Observatory"
  longitude = 100:20.0
  latitude = 35:58.8
  altitude = 212.
  timezone = 6
cl> observ set observatory v+
Using database defined by 'obsdb' environment variable:
/tmp/test/obsdb.dat
Using obs... defined by 'obs...' environment variable: sco
Using observatory parameters for database entry: sco
Observatory parameters for Small College Observatory
  observatory = sco
  timezone = 6
  altitude = 212.
  latitude = 35:58.8
  longitude = 100:20.0
  name = 'Small College Observatory'
```

5. List the observatory assigned to some images with a determined either by the "observatory" environment variable or that set in the observatory task.

```
cl> observ images observatory dev$pix,demoobj1
Observatory parameters for Small College Observatory
  observatory = sco
  timezone = 6
  altitude = 212.
  latitude = 35:58.8
  longitude = 100:20.0
  name = 'Small College Observatory'
Images: dev$pix (default observatory)
Observatory parameters for Kitt Peak National Observatory
  observatory = kpno
```

```
timezone = 7
altitude = 2120.
latitude = 31:58.8
longitude = 111:36.0
name = 'Kitt Peak National Observatory'
Images: demoobj1 (OBSERVAT keyword)
```

6. Set explicit observatory parameters:

```
cl> epar observatory
<set observatory parameters>
cl> observ list obspars
Observatory parameters for North Pole
  observatory = obspars
  timezone = 0
  altitude = 0.
  latitude = 90.
  longitude = 0.
  name = 'North Pole'
```

7. Use observatory parameters in expressions:

```
cl> observ set kpno
Observatory parameters for Kitt Peak National Observatory
  observatory = kpno
  timezone = 7
  altitude = 2120.
  latitude = 31:58.8
  longitude = 111:36.0
  name = 'Kitt Peak National Observatory'
cl> = observ.lat
31.98
cl> = sin (3.14159/180 * observ.lat)
0.52962280742153
```

SEE ALSO

Tasks in astutil, imred, onedspec, and twodspec.